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Department of Civil Engineering | Major project review 2 | Batch 120ABR

Digital twin for live condition monitoring and retrofit decision support of reinforced concrete members

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Aim and objectives

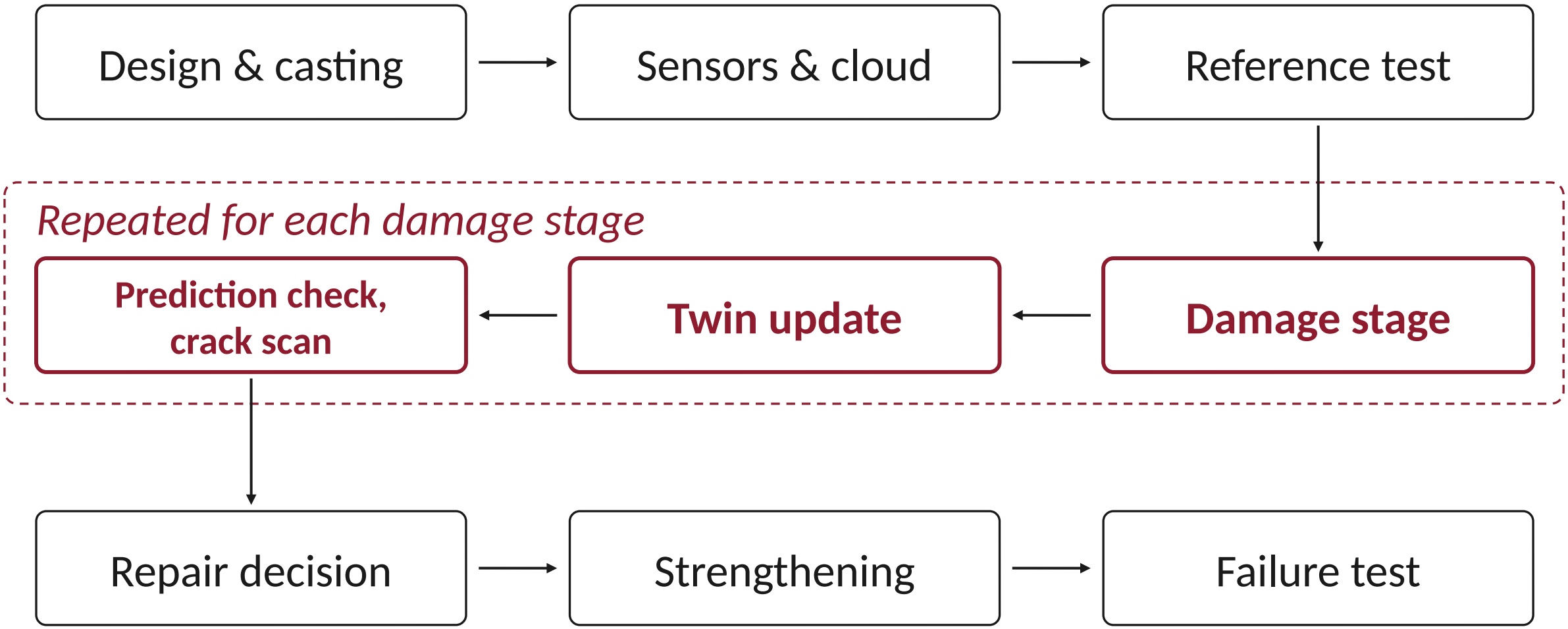
Aim

To develop and experimentally validate a digital twin that measures damage in reinforced concrete beams and recommends retrofitting

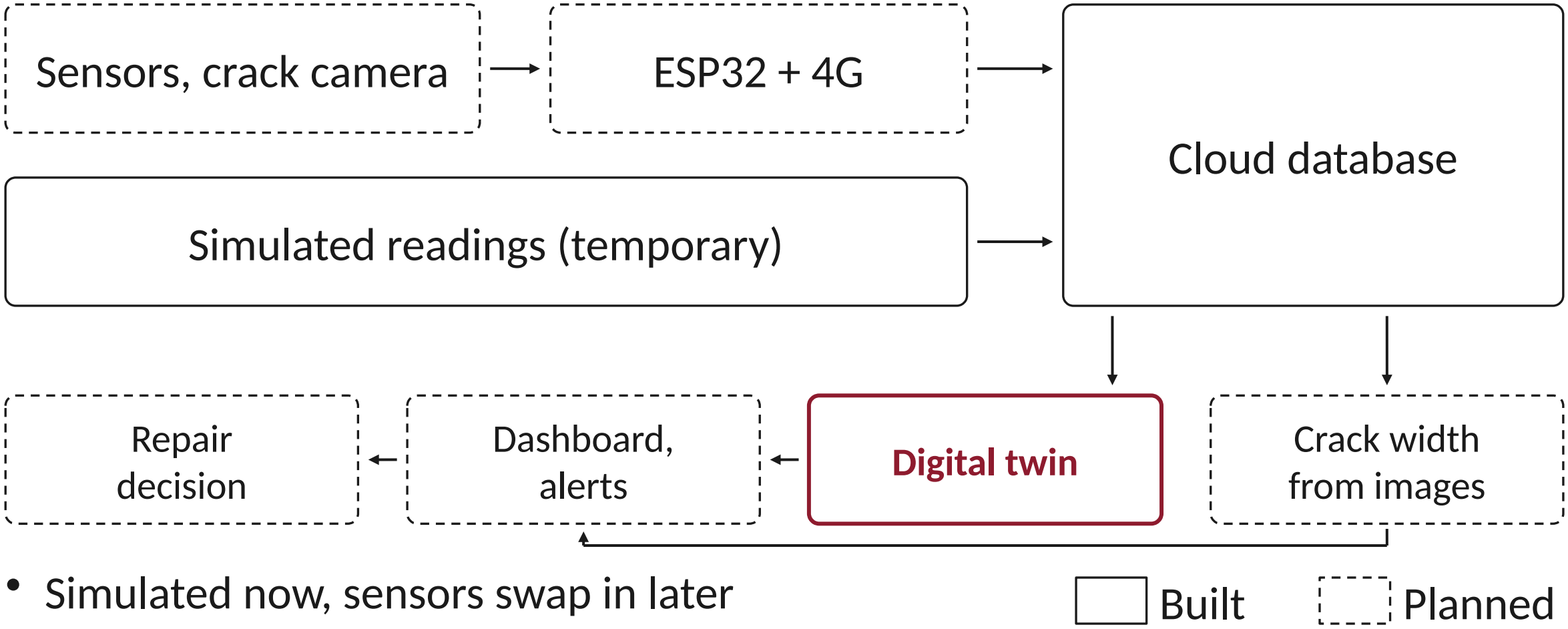
Objectives

1. To design and instrument under-reinforced beams to IS 456:2000 for staged four-point load tests with automated sensing
2. To develop a digital twin that fuses static and vibration data by Bayesian updating to estimate flexural stiffness and predict service deflection
3. To develop a retrofit decision framework from IS 456 checks on simulated beams
4. To validate the twin's predictions and the recommended strengthening by service-load and failure tests

Methodology



Instrumentation and data flow



- Simulated now, sensors swap in later

Specimen details

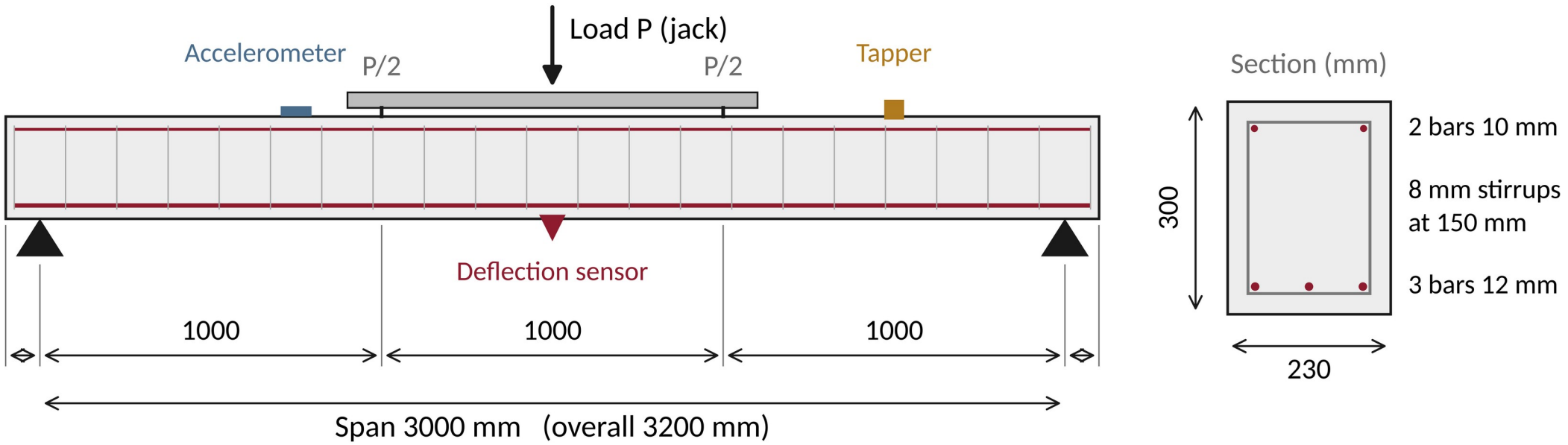


Fig. 1: elevation and cross-section; dimensions in mm

- Concrete M30; main bars Fe500; stirrups Fe415; clear cover 25 mm
- Simply supported, four-point bending; shear span a = 1000 mm

Specimen design (IS 456:2000)

Step	Expression	Result	Clause
Effective depth d	300 − 25 − 8 − 12/2	261 mm	26.4
Steel area A_{st}	$3 \times \pi \times 12^2 / 4$	339 mm²	—
Neutral axis x_u	$0.87 f_y A_{st} / (0.36 f_{ck} b)$	59.4 mm	G-1.1(a)
Limit $x_{u,max}$	0.46 d (Fe500)	120 mm	38.1
Moment capacity M_u	$0.87 f_y A_{st} (d - 0.42 x_u)$	34.8 kN·m	G-1.1(b)
Jack load at M_u	$2 M_u / a$, a = 1.0 m	70 kN	—
Cracking load P_{cr}	$2 f_{cr} I_g / (y_t a)$, $f_{cr} = 0.7 \sqrt{f_{ck}}$	26.5 kN	6.2.2
Shear capacity	$V_c + V_{us} = 31.4 + 63.2$	94.6 kN	40.4
Span / effective depth	3000 / 261	11.5 ≤ 20	23.2.1

- Under-reinforced; shear capacity 2.7 × demand: flexure governs

Model verification and blind validation

Our beam, with bars	STAAD.Pro	OpenSeesPy	Difference
First frequency	50.7 Hz	52.1 Hz	+2.8%
Deflection at 15 kN	0.483 mm	0.470 mm	−2.7%
Shear deformation	included	not included	—

Published beams	Test	Prediction	Difference
Free-free frequency	88.4 Hz	85.5 Hz	−3.3%
Peak load	57.4 kN	55.7 kN	−3.0%
Polymer, Musiał et al. (2021)	90, 116 kN	80, 107 kN	−11%, −8%
Polymer, Ozturkoglu et al. (2024)	52, 66 kN	51, 64 kN	−3%, −3%

- Published inputs used as given: no tuning
- Path from 12 to 45 kN off by up to 29 points

Results: stiffness loss in 300 simulated beams

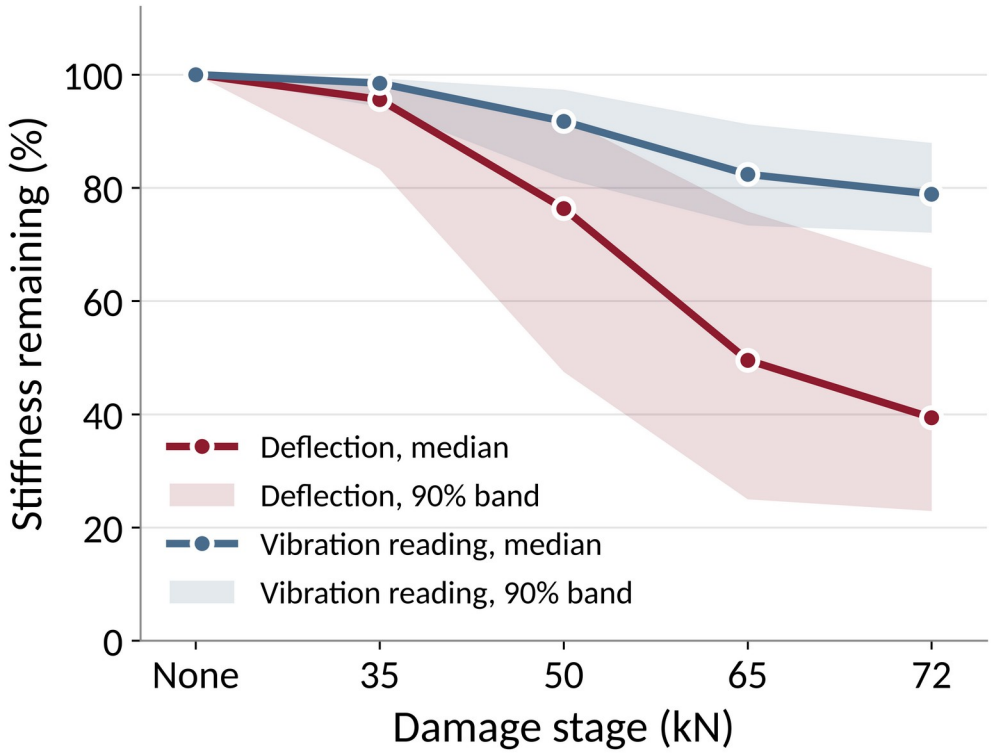


Fig. 2: median and 90% band

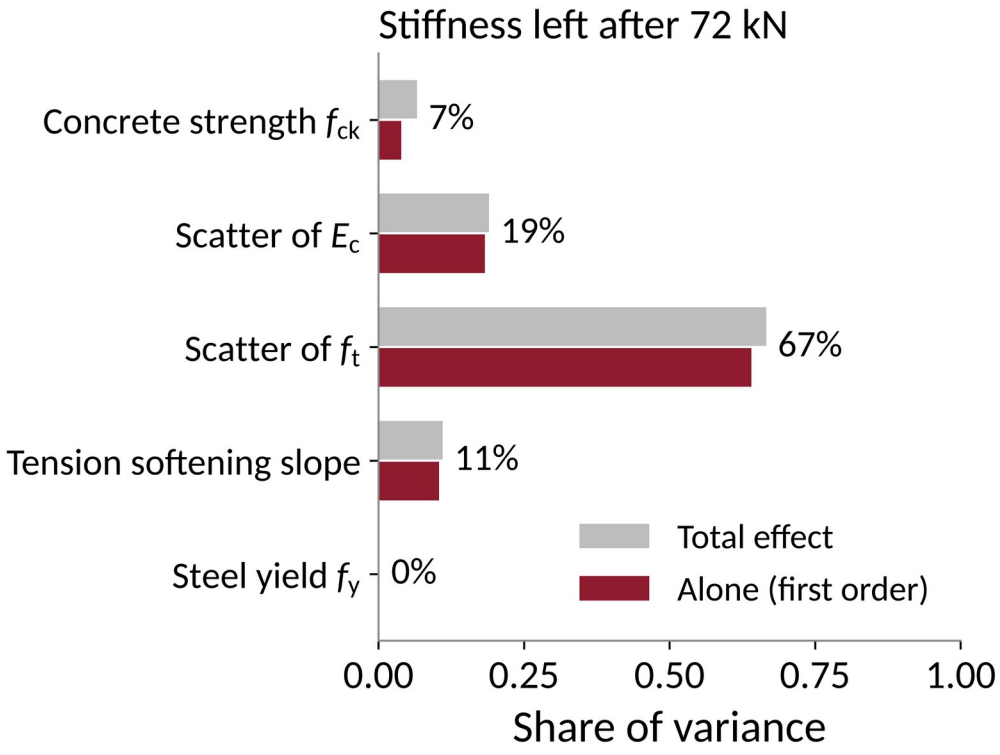
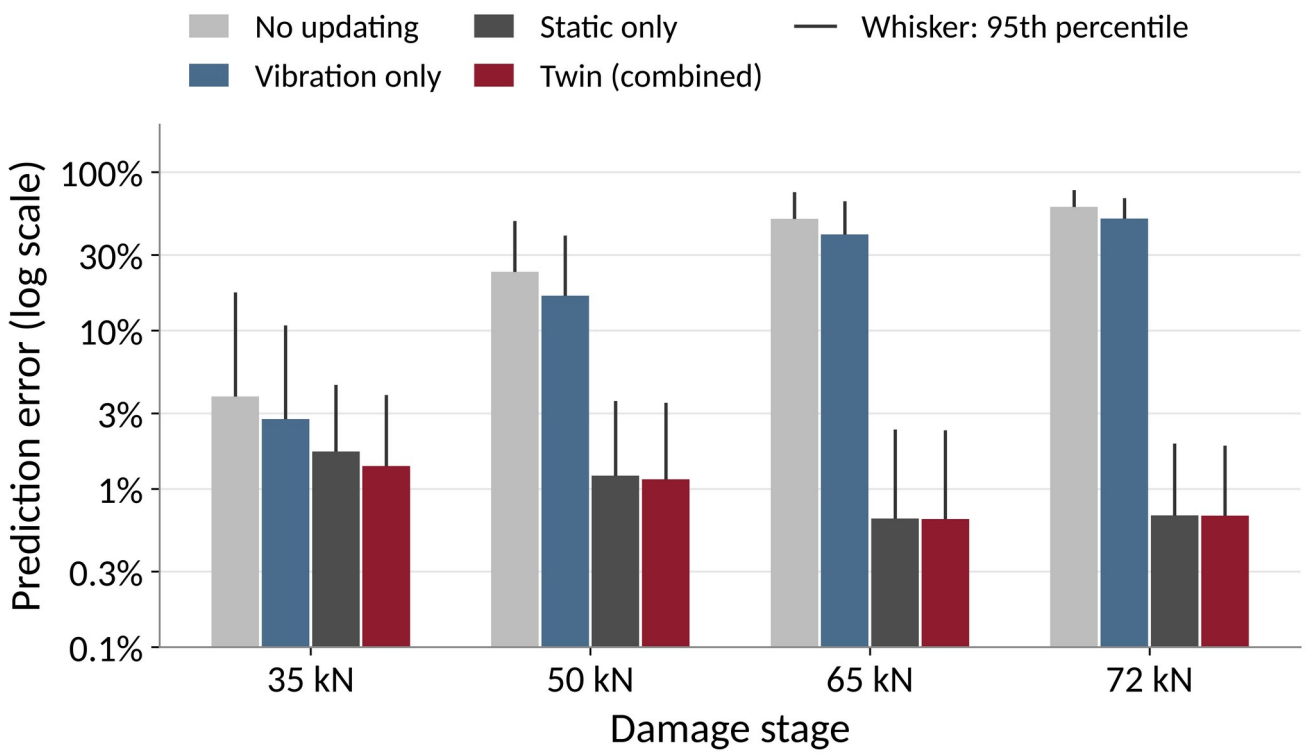


Fig. 3: Sobol sensitivity (R^2 = 0.95)

- Same loads: 23–66% left (simulated), 51% vs 57% (test)

Results: accuracy of the twin's deflection prediction



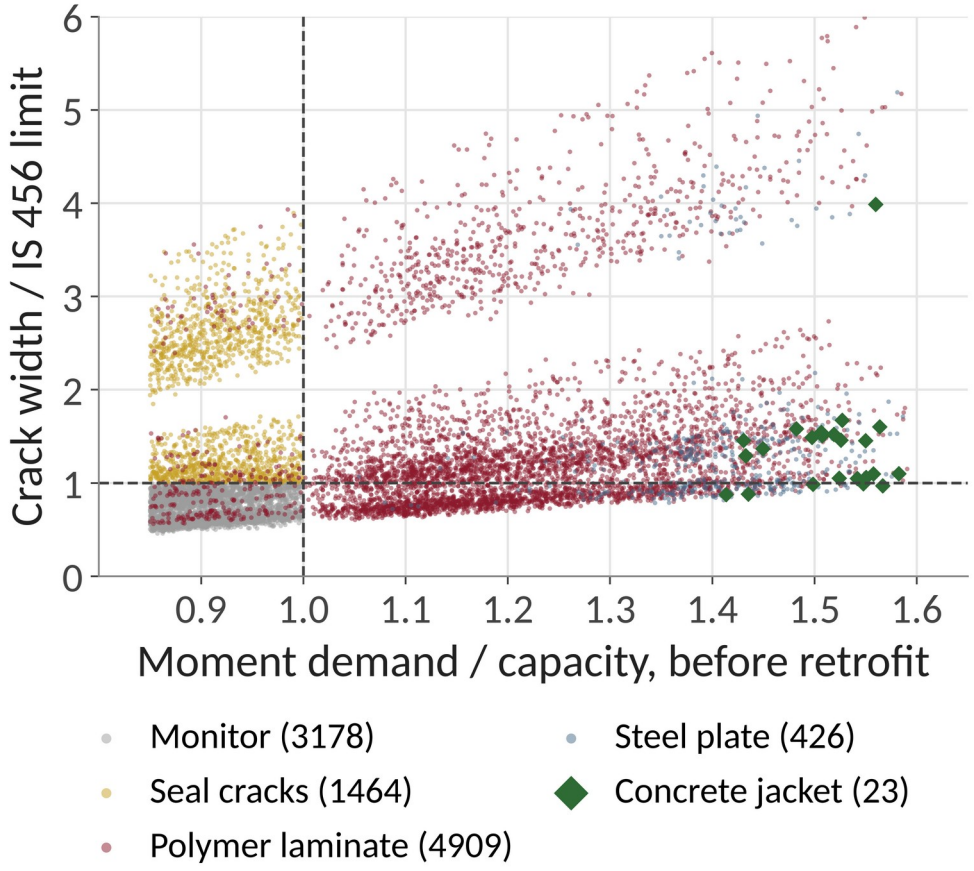
Median error at 72 kN

- No updating: 60%
- Vibration only: 51%
- Twin: 0.7%
- 90% band holds: 81–93%

Fig. 4: prediction error on 150 simulated test beams

- Published test data: learned bias most accurate at every step

Retrofit decision: 10,000 simulated beams



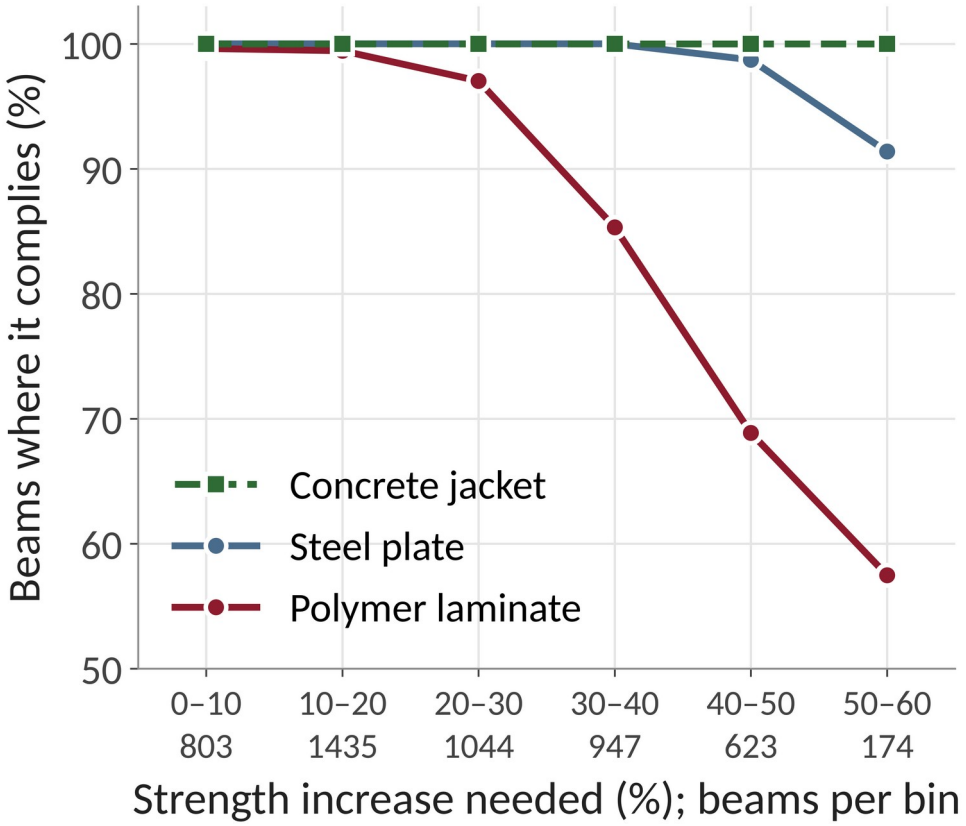


Fig. 5: least intrusive compliant scheme

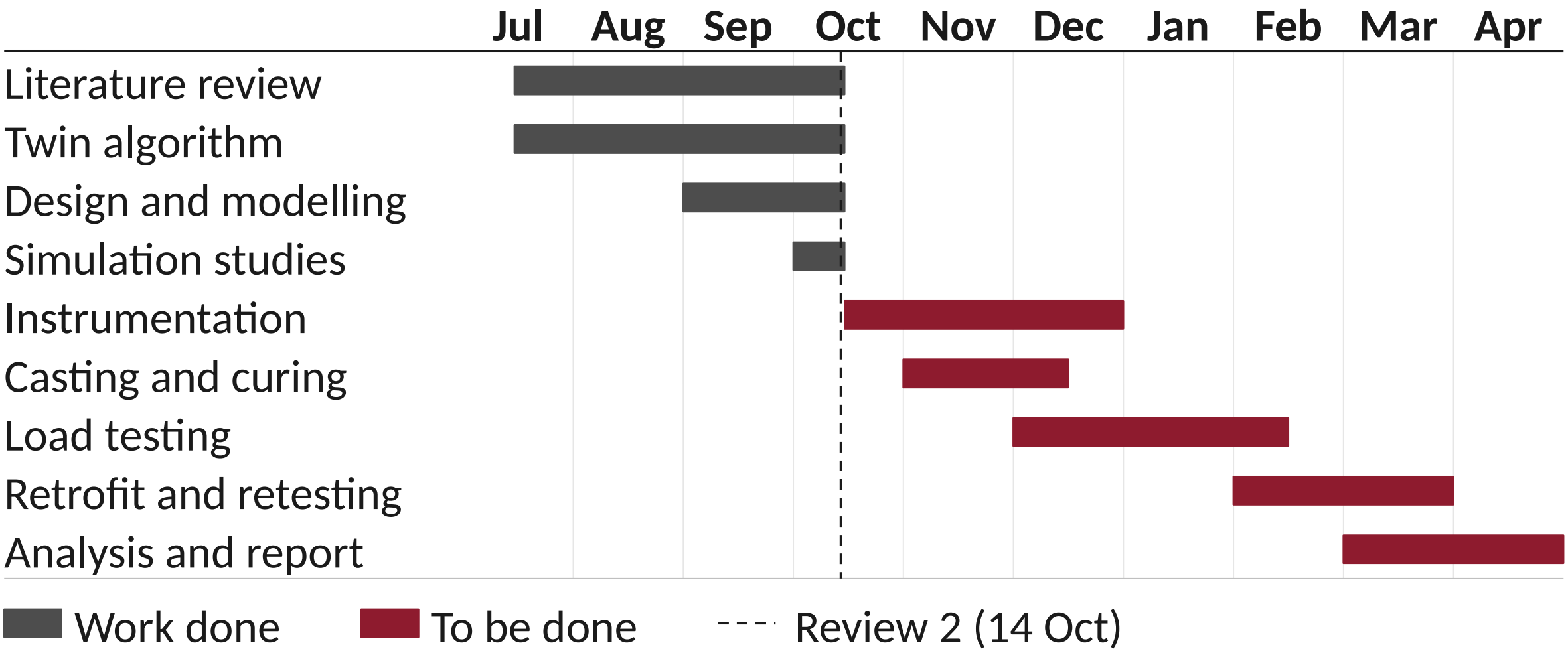
Fig. 6: compliance by strength needed

- Decision tree learned from the simulation: 90% correct on 3000 unseen beams

Summary of work done

- Reviewed the literature on digital twins, damage measurement and strengthening
- Designed the test beam to IS 456, under-reinforced so that flexure governs
- Verified the beam model against hand calculation and STAAD.Pro
- Predicted a published beam test blind, before seeing its results
- Showed by simulation that identical loading damages beams differently
- Showed from published tests that vibration alone underestimates damage
- Developed the digital twin and tested it on simulated and published data
- Checked the polymer strengthening calculation against published beams
- Developed a retrofit decision framework from IS 456 checks on simulated beams

Work done and work to be done (2026–27)



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Thank you